
Mamata Ashoksingh

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OBJECTIVE:

To obtain challenging and responsible position in an organization where I can contribute to the successful growth of an organization using my ability and knowledge. I believe in "There is always a better way of doing things".

SUMMARY:

- Overall **8** years of experience. 7 years in Autosar Development Phase and 1 year worked as integrator.
- Experienced in Kombi Subsystem(KSS) development for instrumentation cluster in CAR multimedia domain.
And Have experience in RADAR domain.
- 4.3 Years of experience in Software Development and Maintenance in **AUTOSAR Service Layer**
(Autosar COM development and Autosar COM to RTE Interfacing). Around 4 years of experience in **RADAR features** like RCTA, RCW, DOA,BSD. Have good knowledge on SQT(software qualification testing) and SIT(Software integration testing).
- Experienced in Embedded Development, designing ,Unit Testing, functional testing in Software Development Life Cycle and familiar with relevant phases of SDLC
- Experienced in development, maintenance and change request implementation of CAN Network module in AUTOSAR architecture.
- Experienced in Message and signal implementation using CAN protocol
- Good at implementing special handlings of signals in case of failures (Timeout, BZ and CRC) during CAN Communication.
- Have Good understanding on Radar features such as blind spot detection (BSD), lane change assistance (LCA), collision Detection (LCDA),and rear cross traffic alert (RCTA) for 2 sensor RR and RL
- Experienced in Root Cause Analysis, Debugging, Issue fixing for Software Maintenance.
- Experienced in Static Code Analysis QAC , Unit Testing, , Offline Trace analysis and Functional Testing.
- Possess Experience in Independent Testing and Pre-Delivery Check for quality assurance of modules.
- Good at code debugging using Green Hills Multi debugger, Lauterbach.
- Experienced in Testing using CANoe and CAPL.
- Quite good at understanding customer requirements and working with **DOORS, Polarion.**

EMPLOYMENT PROFILE:

May, 2018 - July 2022 (Coimbatore,Tamil nadu)	Software Engineer Robert Bosch Engineering and Business Solutions , Coimbatore
July 2022 - till date	Senior Software engineer Aptiv Automotive India Private Company , Bengaluru

SKILLS AND TOOLS EXPERIENCE:

Programming Languages : C, C++ basic, Embedded C

Protocols : CAN, CAN-FD

Frame work : AUTOSAR Architecture

Scripting Languages : Python, CAPL

Development Tools : Eclipse IDE, Microsoft visual studio code, GitHub, Gerrit, Plastic, Lauterbach

Configuration tools : Vector configurator 4.2 SP3, Vector Developer

Requirement and Design Tools : Rational DOORS, Enterprise Architect, Polaron

Testing and debugging tools : Vector CANoe, CAN analyzer, Green Heels, Lauterbach

Review Tools : RADAR, Collaborator, Peer Review Tool

Configuration Management Tool : Rational team Concert (RTC)

Unit Testing : CPP Unit MinGW Code Coverage, Vectorcast

Code quality check : MISRA, QAC, Covarity,

Engineering process : ISO 26262 Functional safety basic Knowledge, ASPICE, Agile, SDLC, functional development, defect analysis, TICS Dashboard.

Projects executed at Robert Bosch Engineering and Business Solutions, Coimbatore (From 2018-2022)

Project Names	1. Audi Variants: MQB, Q3NF, B9, C8, GTetron, A3NF, D5, R8NF Etron, Q2Q3 2. Volkswagen Variants: 536 and 632
Project Summary	Worked in Service layer of AUTOSAR which involves Message and Signal implementation using CAN , interface of Autosar COM with RTE using CAN protocol for Instrument cluster(Car Dashboard)
Tools and Environment	Vector Canoe , GENy, Davinci configurator, Rational Doors , Eclipse RTC , Green Hills Multi, Tera Term, RADAR

Fiat Chrysler Automobiles(FCA)

Project Name	Fiat Chrysler Automobiles(FCA) Variants: Maserati, Alfa965
Project summary	Worked in Base software configuration for Autosar COM
Tools and environment	GIT, GIT Bash, Gerrit, Microsoft visual studio, CANoe
Responsibilities	Change Request Implementation including Requirement Analysis, Implementation, Static Code Analysis QAC, Unit Testing , Functional testing and PDC.

Aptiv Automotive India Private Company , Bengaluru (From 2022 to 2025)

1. Using Plastic for code checkin:

Project Names	TATA Motor Project Variants: PRE-MCE, MCE, MIDSUV TAMOR, MIDSUV CURVV, SIERRA, ROWA HKMC SRR5 CL4 5DR, GN7, CL4 4DR, JX1, MX5
Project Summary	Worked on Application layer , mainly in development activity of RADAR features like BSD, DOA, RCTA, RCW. And SQT(qualification test) and interface testing.
Tools an Environment	Vector Canoe , GENy, Davinci configurator, Rational Doors , Eclipse RTC , Green Hills Multi, Tera Term, RADAR, Polarion, plastic, lauterbach

Aptiv Automotive India Private Company , Bengaluru (From october 2025 to Present)

2. HKMC SRR7

Project Names	HKMC SRR7 Variants : SRR7/FLR7 DN8 PE Proto
Project Summary	• Worked on POC of this HKMC SRR7 • Endtoend ownership of DTC/DID diagnostics and CAN Time Synchronization, with camera ECU acting as global time master
Tools an Environment	Vector Canoe ,Davinci configurator, vector developer, Git, GitHub, Polarion, Lauterbach.

Roles and Responsibilities

- Designed, developed, and integrated **Automotive Communication stacks** including **CAN, CAN FD**, as per **AUTOSAR Classic** and OEM specifications.
- Implemented and maintained **UDS (ISO 14229) diagnostic services** including Read/Write DIDs, DTC handling, Routine Control, Security Access, and ECU Reset.
- Developed and optimized **CAN diagnostics and network monitoring** using **CANoe, CANalyzer**, and **Vector toolchain**.
- Implemented **Time Synchronization mechanisms** such as **CAN Time Sync / Global Time Synchronization** ensuring accurate timestamping across ECUs.
- Worked extensively on **Communication Management (ComM)**, **Network Management (NM)**, and **PDU routing** to support ECU wake-up, sleep, and network modes.
- Integrated and configured **AUTOSAR Basic Software (BSW)** modules including **CanIf, CanSM, Com, PduR, Dem, Dem, EcuM**.
- Implemented fault handling and DTC strategies aligned with OEM diagnostic requirements and functional safety expectations.
- Supported **production issues and field defect analysis**, providing root cause analysis and corrective actions.
- Mentored junior engineers and contributed to **technical reviews and knowledge sharing**.

ACADEMICS:

Course	Board	Institution	Year of passing	Percent age
B.E	VTU (Belgaum)	Appa Institute of Engineering and technology , Kalaburgi	2017	74.22%
XII	State Board Karnataka	Chandrakant Patil memorial PU college of Science, Kalaburgi	2013	80.5%
X	State Board Karnataka	Government Girls High school , Kalaburgi	2011	89.76%

CERTIFICATIONS:

- Obtained certification in C Programming and Core JAVA from Jspider institute
- completed courses in Udemy on CAN protocol and Python
- Attended training in Functional Safety (FuSa) with ISO26262 and Agile process
- Completed certification in AI and Machine learning in Robert Bosch

Achievements:

- Appreciation: Shabash Points awarded for managing complexity in my team.
- Appreciation mail from DH of Department for on time delivery with zero defect

BTech project details :

Project (Team Leader):

The project deals with **Hand Gestures for physically Disabled people**. It will mainly help to communicate with normal people with the hand gesture. The hand gesture will give voice message to interact with people easily.

Extra curriculum activities:

- Took part in Tech-Tanta, Tech Quiz in Avirat fest in the AIET College Kalaburagi.
- Worked as tutor for around 20 students for 4 years.
- Selected for the finals in “Mathematica” Math’s talent examination at school.
- Participated in 3rd International level Science talent search exam at school level (secured 1st class in school level).

Languages known and hobbies:

- languages: Proficient at English, Kannada, Hindi, and beginner in Tamil(Can understand)
- Hobbies: Browsing technical stuff and interested in historical and adventurous movies, Cooking, painting, Writing Thoughts/short poems.

Declaration:

I, Mamata Ashok Singh, hereby declare that the particulars furnished above are true to the best of my knowledge and belief.

Signature

Mamata Ashok Singh